

CDO Mobile — Comprehensive Analytics Tracking Plan

> **Goal:** Instrument the CDO app to measure user behavior, feature adoption, funnel completion, error rates, and business outcomes across all three user roles (ASE, ASM, Outlet).

1. Global / App-Level Events

These events apply across all roles and provide foundational data.

1.1 Session & Lifecycle

Event Name	When to Fire	Key Properties
`app_opened`	App becomes foreground	`platform`, `os_version`, `app_version`, `build_number`, `is_fresh_install`
`app_backgrounded`	App goes to background	`session_duration_ms`, `screen_on_exit`
`app_crashed`	Uncaught exception / ANR	`crash_type`, `screen`, `stack_trace_id`
`session_started`	New session begins (>30s inactivity)	`user_role`, `user_id`, `business_id`, `is_returning_user`
`session_ended`	Session timeout or app kill	`session_duration_ms`, `screens_viewed_count`, `actions_taken_count`
`deep_link_opened`	App opened via deep link	`link_url`, `source`, `campaign`

1.2 Authentication

Event Name	When to Fire	Key Properties
`login_screen_viewed`	Login screen is displayed	`is_auto_redirect` (session expired vs manual)
`login_submitted`	User taps "Send OTP"	`mobile_number_length` (not the number itself)
`login_failed`	OTP request fails	`error_message`, `error_code`
`otp_screen_viewed`	OTP screen loads	`mobile_number_masked`
`otp_entered`	User completes 6-digit OTP	`time_to_enter_ms`
`otp_resend_tapped`	User taps "Resend OTP"	`resend_attempt_number`, `countdown_remaining`
`otp_edit_number_tapped`	User taps "Edit Number"	—
`otp_verified`	OTP successfully verified	`user_role`, `time_to_verify_ms`, `is_first_login`

Event Name	When to Fire	Key Properties
`otp_verification_failed`	OTP verification fails	`error_message`, `attempt_number`
`session_expired`	401 triggers forced logout	`screen_on_expiry`, `session_duration_ms`
`logout_tapped`	User taps logout	`user_role`, `screen`
`logout_confirmed`	Logout completed	`session_duration_ms`

1.3 Navigation & Engagement

Event Name	When to Fire	Key Properties
`screen_viewed`	Any screen becomes visible	`screen_name`, `user_role`, `previous_screen`
`bottom_tab_tapped`	Bottom nav tab tapped	`tab_name`, `previous_tab`, `user_role`
`back_button_tapped`	Hardware or software back	`screen_name`

1.4 Network & Performance

Event Name	When to Fire	Key Properties
`network_status_changed`	Connectivity flips	`new_status` (Available/Unavailable), `screen`
`api_call_completed`	Every API call finishes	`endpoint`, `method`, `status_code`, `latency_ms`, `is_retry`
`api_call_failed`	API call errors out	`endpoint`, `error_type`, `error_message`, `retry_count`
`token_refreshed`	Silent token refresh succeeds	`latency_ms`
`token_refresh_failed`	Token refresh fails (force logout)	`error_code`

2. Role: ASE (Area Sales Executive)

2.1 ASE Home Dashboard

Event Name	When to Fire	Key Properties
`ase_dashboard_loaded`	Dashboard data loads	`total_outlets`, `in_progress`, `asm_pending`, `pending_invoices`, `load_time_ms`
`ase_dashboard_card_tapped`	User taps a stat card	`card_type` (total_outlets / in_progress / asm_pending / pending_invoices)

Event Name	When to Fire	Key Properties
`ase_slab_chart_viewed`	Slab breakdown is visible	`slab_counts` (map of slab → count)

2.2 ASE Outlet List

Event Name	When to Fire	Key Properties
`ase_outlet_list_viewed`	Outlets tab loaded	`outlet_count`, `initial_filter`
`ase_outlet_filter_selected`	Filter chip tapped	`filter_name` (All / Draft / Pending / Active / Onboarded)
`ase_outlet_search_used`	Search input focused + typed	`query_length`, `results_count`
`ase_outlet_card_tapped`	Outlet card tapped	`outlet_id`, `outlet_status`, `action` (view_details / continue_onboarding)
`ase_outlet_list_scrolled`	Scrolled to trigger pagination	`page_loaded`, `total_pages`

2.3 Outlet Onboarding Funnel (■ Critical Business Flow)

> ****This is the most important funnel to track — directly tied to business growth.****

Event Name	When to Fire	Key Properties
`onboarding_started`	FAB "+" tapped or "Continue" tapped	`is_new` (true = fresh, false = resume), `outlet_id` (if resuming), `resume_step`
`onboarding_step_viewed`	Step screen loads	`step_number`, `step_label`, `outlet_id`
`onboarding_step_completed`	Step saved successfully	`step_number`, `step_label`, `time_on_step_ms`, `outlet_id`
`onboarding_step_abandoned`	Back pressed during step	`step_number`, `step_label`, `time_on_step_ms`, `fields_filled_count`
`onboarding_step_error`	Step save API fails	`step_number`, `error_message`, `outlet_id`
`onboarding_completed`	Full 6-step onboarding submitted	`outlet_id`, `total_duration_ms`, `steps_with_errors_count`

Step-specific events:

Event Name	Step	Key Properties
`onboarding_gps_captured`	1 - Basic Details	`lat`, `lng`, `accuracy_meters`, `time_to_capture_ms`
`onboarding_gps_failed`	1 - Basic Details	`error_reason`
`onboarding_pincode_looked_up`	1 - Basic Details	`pincode`, `success`, `auto_filled_city`, `auto_filled_state`

Event Name	Step	Key Properties
`onboarding_outlet_type_selected`	1 - Basic Details	`outlet_type`
`onboarding_slab_selected`	2 - Classification	`slab_id`, `slab_label`
`onboarding_stocking_items_selected`	2 - Classification	`items_count`, `items`
`onboarding_distributor_selected`	3 - Distributor	`distributor_id`, `distributor_name`
`onboarding_bank_details_entered`	3 - Distributor	`has_upi`, `has_cheque_photo`
`onboarding_kyc_type_selected`	4 - KYC	`kyc_id_type`, `has_gst`
`onboarding_kyc_photo_captured`	4 - KYC	`photo_type` (id_proof / gst_certificate / location_proof)
`onboarding_photo_captured`	5 - Photos	`slot_id`, `photo_count`, `total_required`
`onboarding_photo_removed`	5 - Photos	`slot_id`
`onboarding_agreement_accepted`	6 - Agreement	`outlet_id`
`onboarding_agreement_otp_requested`	6 - Agreement	`outlet_id`
`onboarding_agreement_otp_verified`	6 - Agreement	`outlet_id`, `time_to_verify_ms`
`onboarding_submitted`	6 - Agreement	`outlet_id`, `total_duration_ms`

2.4 ASE Outlet Details

Event Name	When to Fire	Key Properties
`ase_outlet_detail_viewed`	Detail screen loaded	`outlet_id`, `outlet_status`, `completion_percent`, `cdo_steps_completed`
`ase_asset_request_tapped`	"Request Asset" tapped	`outlet_id`, `cooler_type`, `capacity`, `signage_type`
`ase_asset_request_success`	Asset request API succeeds	`outlet_id`
`ase_asset_request_failed`	Asset request API fails	`outlet_id`, `error_message`
`ase_compliance_tapped`	"Submit Compliance" tapped	`outlet_id`

2.5 ASE Compliance Submission

Event Name	When to Fire	Key Properties
`compliance_screen_viewed`	Compliance screen loads	`outlet_id`
`compliance_photo_captured`	Compliance photo taken	`slot_id` (coolerImage / signageImage / etc.), `photos_captured_count`
`compliance_photo_removed`	Photo removed	`slot_id`

Event Name	When to Fire	Key Properties
`compliance_submitted`	Submit tapped	`outlet_id`, `cooler_installed`, `signage_installed`, `photos_count`, `time_on_screen_ms`
`compliance_submit_success`	API succeeds	`outlet_id`
`compliance_submit_failed`	API fails	`outlet_id`, `error_message`

2.6 ASE Invoices

Event Name	When to Fire	Key Properties
`ase_invoice_list_viewed`	Invoice tab loaded	`invoice_count`, `initial_filter`
`ase_invoice_filter_selected`	Filter chip tapped	`filter_name`
`ase_invoice_search_used`	Search used	`query_length`, `results_count`
`ase_invoice_detail_viewed`	Invoice detail opened	`invoice_id`, `invoice_status`, `outlet_name`
`ase_invoice_approved`	ASE approves an invoice	`invoice_id`, `has_note`, `time_on_detail_ms`
`ase_invoice_rejected`	ASE rejects an invoice	`invoice_id`, `has_note`, `rejection_reason_length`
`ase_invoice_action_success`	Approve/Reject API succeeds	`invoice_id`, `action` (approve/reject)
`ase_invoice_action_failed`	Approve/Reject API fails	`invoice_id`, `action`, `error_message`

3. Role: ASM (Area Sales Manager)

3.1 ASM Home Dashboard

Event Name	When to Fire	Key Properties
`asm_dashboard_loaded`	Dashboard data loads	`total_ases`, `total_outlets`, `active_outlets`, `in_progress`, `suspended`, `asm_pending`, `pending_invoices`, `load_time_ms`
`asm_dashboard_card_tapped`	Stat card tapped	`card_type` (total_ases / total_outlets / active / in_progress / suspended / asm_pending / pending_invoices)
`asm_my_team_tapped`	"My Team" link tapped	—

3.2 ASM Outlet List & Review

Event Name	When to Fire	Key Properties
`asm_outlet_list_viewed`	Outlets tab loaded	`outlet_count`, `initial_filter`
`asm_outlet_filter_selected`	Filter chip tapped	`filter_name`
`asm_outlet_review_viewed`	Outlet review screen opened	`outlet_id`, `outlet_status`, `onboarded_by_ase`
`asm_outlet_approved`	ASM approves an outlet	`outlet_id`, `time_on_review_ms`
`asm_outlet_rejected`	ASM rejects an outlet	`outlet_id`, `rejection_reason`, `time_on_review_ms`

3.3 ASM Invoices

Event Name	When to Fire	Key Properties
`asm_invoice_list_viewed`	Invoice tab loaded	`invoice_count`, `initial_filter`
`asm_invoice_filter_selected`	Filter chip tapped	`filter_name`
`asm_invoice_detail_viewed`	Invoice detail opened	`invoice_id`, `invoice_status`, `submitted_by_ase`
`asm_invoice_approved`	ASM approves an invoice	`invoice_id`, `has_note`
`asm_invoice_rejected`	ASM rejects an invoice	`invoice_id`, `has_note`

3.4 ASM My Team

Event Name	When to Fire	Key Properties
`asm_my_team_viewed`	My Team screen loaded	`ase_count`
`asm_my_team_searched`	Search used	`query_length`, `results_count`
`asm_my_team_refreshed`	Pull-to-refresh triggered	`ase_count_after_refresh`

4. Role: Outlet (Retailer)

4.1 Outlet Walkthrough (First-Time Onboarding) (■ Critical Conversion Funnel)

Event Name	When to Fire	Key Properties
`walkthrough_started`	Welcome screen loads	`is_first_time`
`walkthrough_welcome_completed`	Next tapped on Welcome	`time_on_step_ms`
`walkthrough_outlet_details_viewed`	Outlet Details step loads	—
`walkthrough_outlet_details_completed`	Outlet details saved	`time_on_step_ms`
`walkthrough_agreement_viewed`	Agreement step loads	—

Event Name	When to Fire	Key Properties
`walkthrough_agreement_completed`	Agreement accepted	`time_on_step_ms`
`walkthrough_payment_viewed`	Payment summary loads	—
`walkthrough_completed`	Final step completed → Dashboard	`total_duration_ms`, `steps_completed`
`walkthrough_abandoned`	User backs out during walkthrough	`step_on_exit`, `time_spent_ms`

4.2 Outlet Dashboard

Event Name	When to Fire	Key Properties
`outlet_dashboard_loaded`	Dashboard data loads	`tier`, `total_invoices`, `pending_invoices`, `last_payout_amount`, `monthly_performance`, `load_time_ms`
`outlet_dashboard_card_tapped`	Stat card tapped	`card_type` (total_invoices / pending_invoices / upload_invoice / view_all_invoices)
`outlet_notification_viewed`	Notification visible in list	`notification_id`, `notification_type`
`outlet_notification_dismissed`	Notification dismissed	`notification_id`, `notification_type`
`outlet_upload_invoice_tapped`	"Upload Invoice" CTA tapped	`source` (dashboard_card / dashboard_button)
`outlet_recent_invoice_tapped`	Recent invoice tapped	`invoice_id`, `invoice_status`

4.3 Invoice Upload Funnel (■ Critical Business Flow)

> ****This funnel directly drives revenue — priority #1 for optimization.****

Event Name	When to Fire	Key Properties
`invoice_upload_started`	Upload Invoice screen loads	`source` (dashboard / invoice_list)
`invoice_capture_method_selected`	Scan or Gallery chosen	`method` (camera / gallery)
`invoice_image_captured`	Image captured/selected	`method`, `capture_time_ms`
`invoice_crop_review_viewed`	Crop & rotate screen shown	—
`invoice_crop_confirmed`	User confirms cropped image	`time_on_crop_ms`
`invoice_crop_retake`	User taps "Retake"	`retake_count`
`invoice_preprocessing_started`	Preprocessing pipeline begins	—
`invoice_preprocessing_completed`	Pipeline finishes	`duration_ms`, `quality_issues_count`, `has_hard_block`
`invoice_quality_warning_shown`	Soft quality warning displayed	`issues` (list of issue types)
`invoice_quality_warning_dismissed`	User proceeds despite warning	`issues`
`invoice_retake_guide_shown`	Hard block guide displayed	—

Event Name	When to Fire	Key Properties
`invoice_retake_guide_dismissed`	User dismisses guide	–
`invoice_filter_selected`	Visual filter changed	`filter_name` (grayscale / original / enhanced)
`invoice_ai_extraction_started`	AI extraction begins	–
`invoice_ai_extraction_phase_changed`	Phase changes (uploading → error)	`phase`
`invoice_ai_extraction_completed`	AI extraction succeeds	`duration_ms`, `items_extracted`, `confidence_avg`
`invoice_ai_extraction_failed`	AI extraction fails	`reason`, `retry_count`, `duration_ms`
`invoice_ai_extraction_cancelled`	User cancels extraction	`phase_on_cancel`, `time_elapsed_ms`
`invoice_review_viewed`	Review screen loads	`items_count`, `grand_total`, `is_auto_filled`
`invoice_review_field_edited`	User edits a pre-filled field	`field_name`, `was_auto_filled`
`invoice_review_line_item_edited`	Line item field edited	`item_index`, `field_name` (product / quantity / price / etc)
`invoice_review_line_item_added`	Manual line item added	`current_item_count`
`invoice_review_line_item_removed`	Line item removed	`item_index`
`invoice_region_scan_used`	ROI manual OCR used	`item_index`, `field_name`
`invoice_submitted`	Final submit tapped	`items_count`, `grand_total`, `was_auto_filled`, `edits_made_count`, `time_on_review_ms`
`invoice_submit_success`	API succeeds	`invoice_id`, `total_upload_duration_ms`
`invoice_submit_failed`	API fails	`error_message`, `retry_count`
`invoice_draft_saved`	Draft saved	`items_count`

4.4 Outlet Invoice List

Event Name	When to Fire	Key Properties
`outlet_invoice_list_viewed`	Invoice tab loaded	`invoice_count`, `initial_filter`
`outlet_invoice_filter_selected`	Filter chip tapped	`filter_name` (All / Submitted / Approved / Rejected / Paid)
`outlet_invoice_search_used`	Search used	`query_length`, `results_count`
`outlet_invoice_detail_viewed`	Invoice detail opened	`invoice_id`, `invoice_status`, `invoice_amount`
`outlet_invoice_list_scrolled`	Pagination triggered	`page_loaded`, `total_pages`

5. Profile & Settings

Event Name	When to Fire	Key Properties
`profile_viewed`	Profile tab opened	`user_role`, `has_kyc_data`
`profile_my_team_tapped`	"My Team" tapped (ASM only)	–

6. Funnel Definitions (for Dashboards)

These are the key funnels to visualize in your analytics dashboard.

6.1 Authentication Funnel

...

login_screen_viewed → login_submitted → otp_screen_viewed → otp_entered → otp_verified

...

****KPIs:**** Conversion rate, drop-off at each step, avg time to verify, OTP resend rate

6.2 Outlet Onboarding Funnel (ASE)

...

onboarding_started

→ Step 1: Basic Details (onboarding_step_completed, step=1)

→ Step 2: Classification & Volume (step=2)

→ Step 3: Distributor & Bank (step=3)

→ Step 4: KYC Details (step=4)

→ Step 5: Photos & Verification (step=5)

→ Step 6: PFP Agreement (step=6)

→ onboarding_completed

...

****KPIs:****

- Step-by-step drop-off rate

- Average time per step

- Resume rate (how often users return to finish draft outlets)

- Completion rate (started → completed)

- Most common abandonment step

- GPS capture failure rate

6.3 Outlet Approval Funnel (ASM)

...

asm_outlet_review_viewed → asm_outlet_approved / asm_outlet_rejected

...

****KPIs:**** Approval rate, rejection reasons, avg review time

6.4 Invoice Upload Funnel (Outlet)

...

invoice_upload_started

→ invoice_image_captured

→ invoice_crop_confirmed

→ invoice_ai_extraction_started

→ invoice_ai_extraction_completed

→ invoice_review_viewed

→ invoice_submitted

→ invoice_submit_success

...

****KPIs:****

- Capture-to-submit conversion rate
- AI extraction success rate
- Average edits per extracted invoice (auto-fill quality signal)
- Quality gate block rate (retake rate)
- Time from capture to successful submission
- Draft save rate

6.5 Invoice Review Funnel (ASE → ASM)

...

ase_invoice_detail_viewed → ase_invoice_approved / ase_invoice_rejected

asm_invoice_detail_viewed → asm_invoice_approved / asm_invoice_rejected

...

****KPIs:**** Approval rate per level, rejection reasons, avg review time

6.6 Retailer Walkthrough Funnel

...

walkthrough_started → walkthrough_welcome_completed → walkthrough_outlet_details_completed →
walkthrough_agreement_completed → walkthrough_completed

...

KPIs: Step-by-step conversion, abandonment rate, total completion time

6.7 CDO Lifecycle Funnel (End-to-End)

...

Enrollment → ASM Approval → Asset Request → CDO Verification → Active Outlet

...

KPIs: Time in each stage, bottleneck stages, outlets stuck in each status

7. User Properties (Super Properties)

Set these once per session or on login — they are attached to every subsequent event.

Property	Description
`user_id`	Unique user identifier
`user_role`	ASE / ASM / OUTLET
`business_id`	Business the user belongs to
`outlet_id`	(Outlet only) The current outlet's ID
`distributor_id`	(Outlet only) Linked distributor ID
`app_version`	Semantic version (e.g., 1.2.3)
`build_number`	Build number
`platform`	iOS / Android
`os_version`	OS version string
`device_model`	Device model name
`network_type`	WiFi / Cellular / None
`locale`	Device locale

8. Recommended Analytics SDK & Architecture

SDK Options for KMP (Compose Multiplatform)

- Firebase Analytics** — Best for mobile-first, free, native SDKs with `expect/actual` wrappers

- 2. **Mixpanel** — Better funnel analysis, cohort analysis, and retention tracking
- 3. **Amplitude** — Best for product analytics, journey maps, and behavioral cohorts
- 4. **PostHog** — Open-source option with session replay

Suggested Architecture Pattern

```
...

AnalyticsTracker ← commonMain interface

fun track(event: AnalyticsEvent)
fun setUserProperty(key, value)
fun identify(userId)
fun reset()

Android Impl iOS Impl ← expect/actual
(Firebase/ (Firebase/
Mixpanel) Mixpanel)

...
```

Event Naming Convention

- **snake_case** for event names (e.g., `invoice_upload_started`)
- **Prefix by feature area** (e.g., `ase_`, `asm_`, `outlet_`, `onboarding_`)
- **Suffix with outcome** for result events (e.g., `_success`, `_failed`, `_completed`)
- **Properties use snake_case** (e.g., `time_on_step_ms`, `outlet_id`)

9. Priority & Phasing

Phase 1 — Must Have (Launch Critical)

Priority	Events
P0	Authentication funnel (login → OTP → verified)

Priority	Events
P0	Screen views (all screens)
P0	Onboarding funnel (start → each step → complete)
P0	Invoice upload funnel (capture → extract → review → submit)
P0	App open / session tracking
P0	Error tracking (API failures, crashes)

Phase 2 — High Value (Week 2-3) ■

Priority	Events
P1	Invoice approve/reject actions (ASE + ASM)
P1	Outlet approval funnel (ASM)
P1	Dashboard engagement (card taps, navigation patterns)
P1	Walkthrough funnel (Outlet first-time)
P1	Compliance submission funnel
P1	Network status and API latency tracking

Phase 3 — Optimization (Month 2+) ■

Priority	Events
P2	Search usage analytics
P2	Filter usage patterns
P2	Notification engagement
P2	Pagination / scroll depth
P2	AI extraction quality metrics (edits per invoice)
P2	My Team engagement (ASM)
P2	Session recording / heatmaps

10. Key Business Metrics to Derive

Metric	Formula	Business Value
Onboarding Completion Rate	`onboarding_completed` / `onboarding_started`	Outlet acquisition efficiency
Avg Onboarding Time	`avg(onboarding_completed.total_dur ation_ms)`	Field ops efficiency

Metric	Formula	Business Value
Most Dropped Step	Step with highest `step_abandoned` count	UX improvement target
Invoice Auto-Fill Quality	`avg(edits_per_extracted_invoice)`	AI extraction accuracy
Invoice Upload Success Rate	`invoice_submit_success / invoice_upload_started`	Revenue funnel health
Approval Turnaround Time	Time between `submitted` and `approved` events	Operational efficiency
Rejection Rate	`rejected / (approved + rejected)` per role	Quality signal
DAU/MAU Ratio	Daily Active / Monthly Active users	Stickiness
Feature Adoption	% of users using each feature	Product-market fit
Session Frequency	Sessions per user per week by role	Engagement health
GPS Failure Rate	`gps_failed / (gps_captured + gps_failed)`	Device/field issue signal
Image Quality Block Rate	`retake_guide_shown / image_captured`	Camera UX quality
Walkthrough Completion Rate	`walkthrough_completed / walkthrough_started`	Retailer activation
CDO Lifecycle Velocity	Avg days from Enrollment → Onboarded	Go-to-market speed